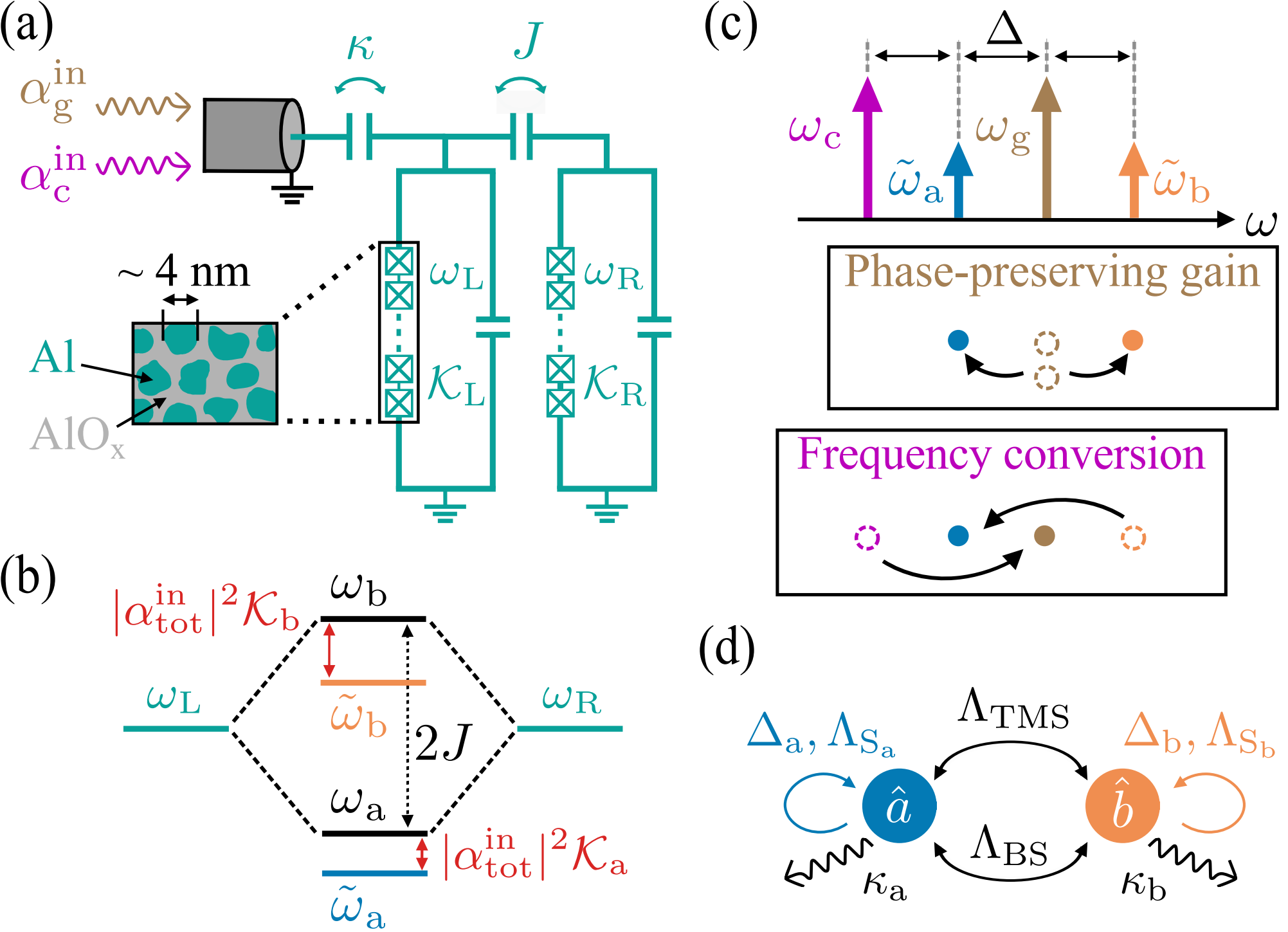
(a)
α_g^{in}
α_c^{in}
κ
J
~ 4 nm
Al
AlO_x
ω_L
$\mathcal{K}_L$
ω_R
$\mathcal{K}_R$
(b)
$|\alpha_{tot}^{in}|^2\mathcal{K}_b$
ω_b
ω_L
$\tilde{\omega}_b$
$2J$
ω_R
ω_a
$|\alpha_{tot}^{in}|^2\mathcal{K}_a$
$\tilde{\omega}_a$
(c)
Δ
ω_c
$\tilde{\omega}_a$
ω_g
$\tilde{\omega}_b$
ω
Phase-preserving gain
Frequency conversion
(d)
Δ_a, Λ_{S_a}
Λ_{TMS}
Δ_b, Λ_{S_b}
$\hat{a}$
$\hat{b}$
Λ_{BS}
κ_a
κ_b